Quantifying Network Similarity using Graph Cumulants[1]

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$\mathcal{TL;DR}$: Graph cumulants perform better and are more intuitive than the typical subgraph statistics[3]

Using only moments is **awkward**[4]

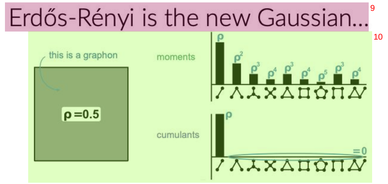
"The length of a human averages 1.7 meters,
and their **average squared length** is 2.9 square meters"[5]

Using cumulants is **easier** to understand[6]

"The length of a human has: a **variance** of 0.01 meters,
a **standard deviation** of 0.1 meters,
a **relative fluctuation** of 6%"[7]

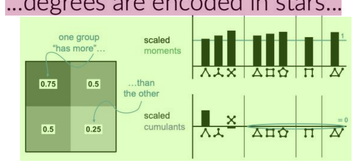
Graph Cumulants: the Better Subgraph Statistics[8]

Erdős-Rényi is the new Gaussian...[9]



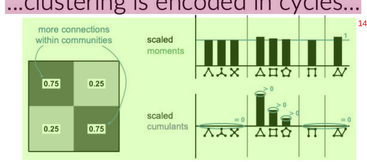
[10]

...degrees are encoded in stars...[11]

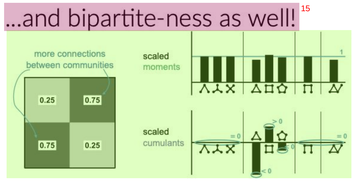


[12] [13]

...clustering is encoded in cycles...[14]



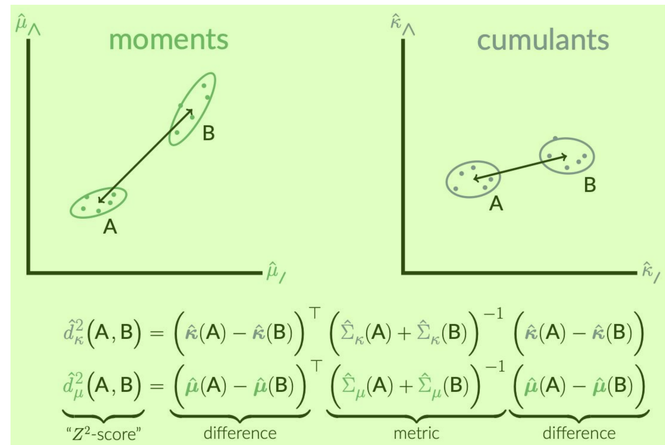
...and bipartite-ness as well![15]



[16]

But are graph cumulants better for testing?[17]
(than subgraph densities)

Apples-to-Apples: A Two-Sample Test[18]



[19]

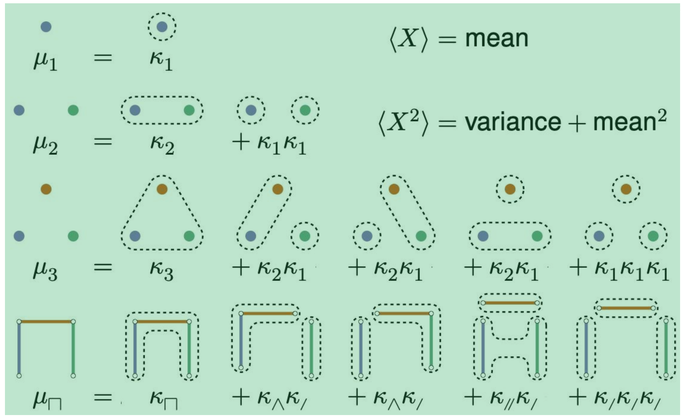
When estimating the covariance...[20]

$$\mathrm{Cov}(\hat{\mu}_g, \hat{\mu}_{g'}) = \underbrace{\langle \hat{\mu}_g\, \hat{\mu}_{g'} \rangle}_{\text{"hard"}} - \underbrace{\langle \hat{\mu}_g \rangle \langle \hat{\mu}_{g'} \rangle}_{\text{"easy"}}$$ [21]

...the "hard" part uses a combinatorial disjoint union rule[22]

$$c_{\wedge} c_{/} = 4c_{\wedge} + 2c_{\Delta} + 2c_{\curlywedge} + 4c_{\sqcap} + c_{/\!\wedge}$$ [23]

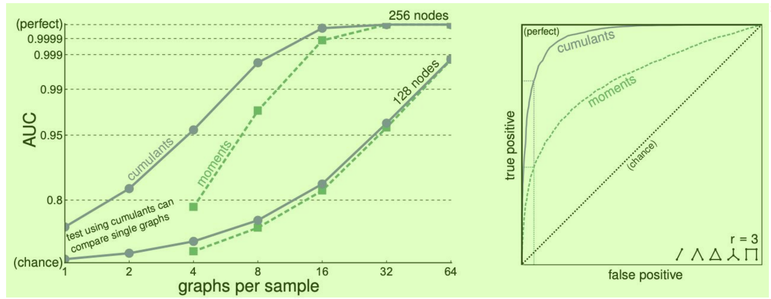
Combinatorial Construction of Cumulants[24]



[25]

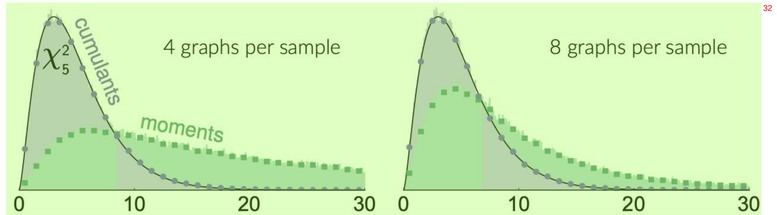
Graph Cumulants Clearly Conquer[26]

Graph cumulants outperform subgraph densities in general...[27]
...and graph cumulants also work for single graph samples![28]



[29]

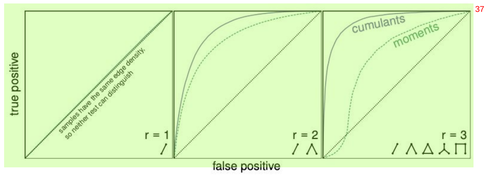
Why do graph cumulants perform better?[30]
...because their **fluctuations** look more $\mathcal{N}$ormal![31]



[32]

"Z^2-score" for graphs sampled from the same distribution[33]

Graph Cumulants in the (semi-)Wild![34]



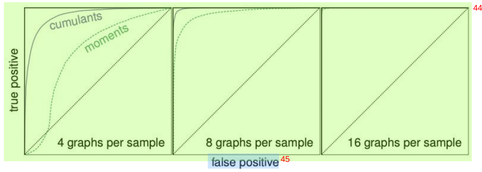
[37]

Varying:[35]
Number of subgraphs[36] used by both tests

Comparing:[38]
Genetic interaction networks[39] of Arabidopsis and Mouse

Varying:[40]
Number of graphs per sample[41] used by both tests

Comparing:[42]
Genetic interaction networks[43] of Human and Rat



[44] [45]